

Lab 3: Component Modelling & Architectural Pattern Selection

Virtual Technical Conference Platform

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Architecture Selection

We chose the Microservices Architecture for the Virtual Technical Conference Platform System.

Reason 1 — Independent Scaling for Concurrent Load

NFR-001 requires the platform to support thousands of simultaneous attendees and concurrent virtual sessions without degradation. Microservices allow the Session Scheduling Service and Notification Service to be scaled out independently during peak conference hours (e.g., session start times), while lighter-traffic services such as Proposal Submission remain at baseline capacity, avoiding the cost and complexity of scaling an entire monolith.

Reason 2 — Independent Development and Fault Isolation Across Workflows

The platform's functional requirements (FR-001 to FR-005) map to distinct, loosely-coupled workflows: proposal submission, review/decision, notification, scheduling, and file uploads. Microservices let each workflow evolve, deploy, and fail independently — for example, a fault in the Notification Service (e.g., an email provider outage) does not block Track Chairs from making acceptance decisions or authors from submitting proposals, improving overall availability.

Security Advantage

NFR-002 requires encrypted data in transit/at rest and malware-scanned file uploads. Isolating the File Storage Service as its own microservice lets it enforce a dedicated security boundary — encryption at rest, TLS in transit, and malware scanning — as a single hardened entry point for all uploaded files (proposals and camera-ready materials), rather than distributing file-handling logic (and its attack surface) across every service.

Performance Benefit

Because each microservice owns its own data and scaling policy, the Review & Decision Service (subject to bursty load when Track Chairs process many submissions near deadlines) can be scaled or cached independently of the Session Scheduling Service, reducing response latency for authors checking submission status without over-provisioning the entire system.

Component Diagram

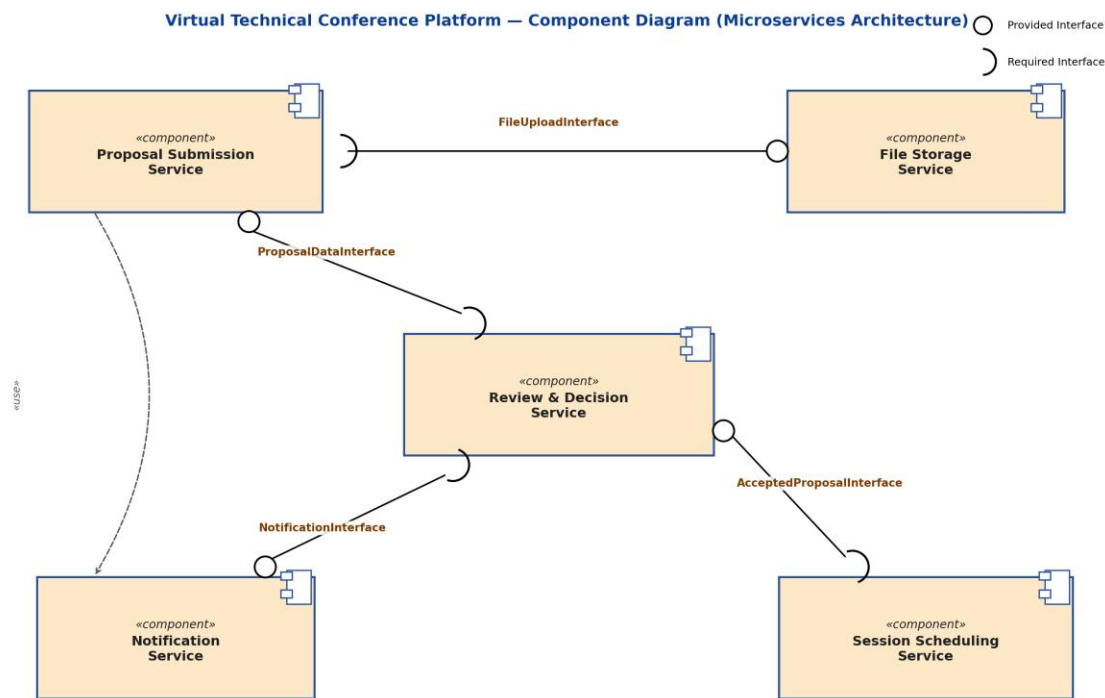


Figure 1: Microservices-based component diagram for the Virtual Technical Conference Platform, showing 5 components and 4 assembly-connected provided/required interfaces.